Date: 13th July 2026

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Course Code and Name: 2CS502CC26 Object-Oriented Programming

Practical: 2

Practical 2(a)

Aim: Write a program to determine whether a year is a leap year or not.

Methodology followed:

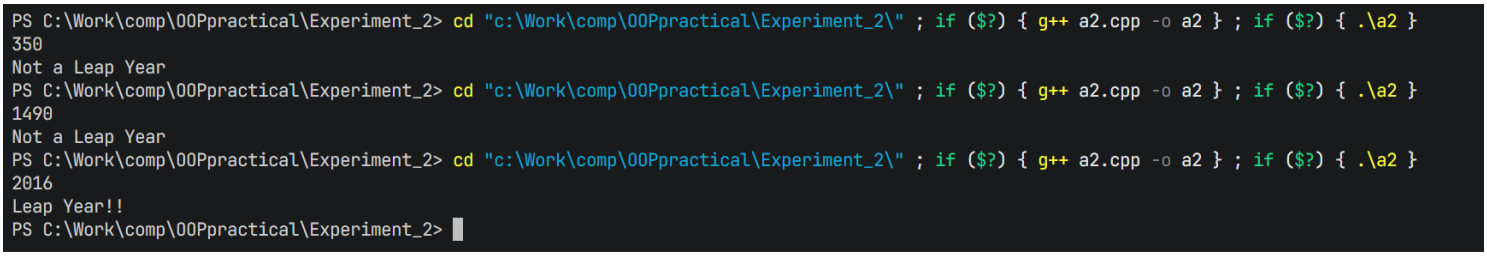
```
#include <iostream>
using namespace std;

int main()
{
    int year;
    cin >> year;

    // Leap year if divisible by 4
    if (year % 4 == 0)
    {
        cout << "Leap Year!!" << endl;
    }
```

```
    else
    {
        cout << "Not a Leap Year" << endl;
    }
}
```

Input and Output:



Practical 2(b)

Aim: Write a menu-driven program to calculate the area of different shapes using switch-case.

Methodology followed:

```
#include <iostream>
using namespace std;

int main()
{
    cout << "Enter 0 for Circle , 1 for rectangle , 2 for square , 3 for traingle: " << endl;
    int choice;
    cin >> choice;

    switch (choice)
    {
    case 0: // Circle: Area = π × r²
        cout << "Enter the radius: " << endl;
        int radius;
        cin >> radius;
```

```
        cout << "The area of the circle with radius "
<< radius << " is " << 3.14 * radius * radius;
        break;

    case 1: // Rectangle: Area = length × width
        cout << "Enter the side 1 and side 2: " <<
endl;
        int a, b;
        cin >> a >> b;
        cout << "The area of the rectangle with sides "
<< a << " and " << b << " is " << a * b;
        break;

    case 2: // Square: Area = side × side
        cout << "Enter the side length: " << endl;
        cin >> a;
        cout << "The area of the square with side " <<
a << " is " << a * a;
        break;

    case 3: // Triangle: Area = 0.5 × base × height
        cout << "Enter the base and height: " << endl;
        cin >> a >> b;
```

```
        cout << "The area of the traingle with base "
<< a << " and height " << b << " is " << 0.5 * a * b;
        break;
    }
}
```

Input and Output:

```
PS C:\Work\comp\OOPpractical\Experiment_2> cd "c:\Work\comp\OOPpractical\Experiment_2\" ; if ($?) { g++ b2.cpp -o b2 } ;
if ($?) { .\b2 }
Enter 0 for Circle , 1 for rectangle , 2 for square , 3 for traingle:
0
Enter the radius:
12
The area of the circle with radius 12 is 452.16
PS C:\Work\comp\OOPpractical\Experiment_2> cd "c:\Work\comp\OOPpractical\Experiment_2\" ; if ($?) { g++ b2.cpp -o b2 } ;
if ($?) { .\b2 }
Enter 0 for Circle , 1 for rectangle , 2 for square , 3 for traingle:
1
Enter the side 1 and side 2:
12 45
The area of the rectangle with sides 12 and 45 is 540
PS C:\Work\comp\OOPpractical\Experiment_2> cd "c:\Work\comp\OOPpractical\Experiment_2\" ; if ($?) { g++ b2.cpp -o b2 } ;
if ($?) { .\b2 }
Enter 0 for Circle , 1 for rectangle , 2 for square , 3 for traingle:
2
Enter the side length:
14
The area of the square with side 14 is 196
PS C:\Work\comp\OOPpractical\Experiment_2> cd "c:\Work\comp\OOPpractical\Experiment_2\" ; if ($?) { g++ b2.cpp -o b2 } ;
if ($?) { .\b2 }
Enter 0 for Circle , 1 for rectangle , 2 for square , 3 for traingle:
3
Enter the base and height:
19 5
The area of the traingle with base 19 and height 5 is 47.5
PS C:\Work\comp\OOPpractical\Experiment_2>
```

Practical 2(c)

Aim: Write a program to display the Fibonacci series up to N terms.

Methodology followed:

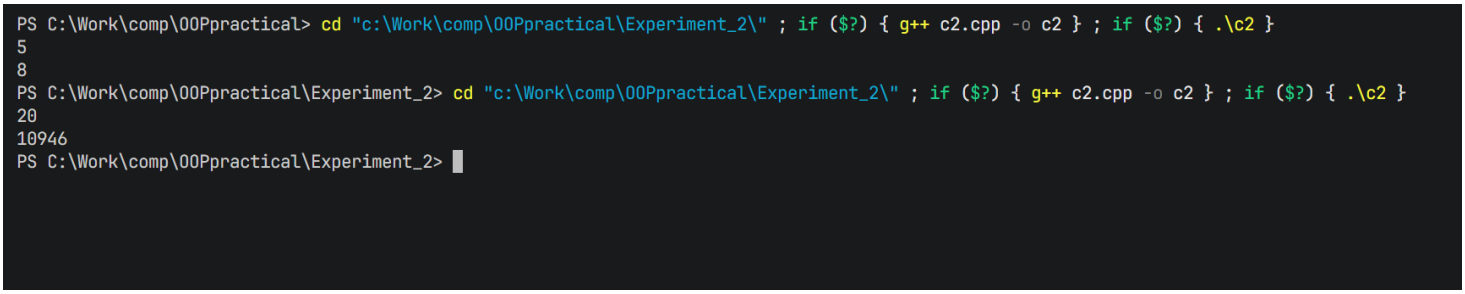
```
#include <iostream>
using namespace std;

// Recursive function: F(n) = F(n-1) + F(n-2)
int f(int n)
{
    if (n == 1 || n == 0)
    {
        return 1;
    }
    else
    {
        return f(n - 2) + f(n - 1);
    }
}

int main()
{
```

```
    int num;
    cin >> num;
    cout << f(num);
}
```

Input and Output:



Practical 2(d)

Aim: Write a program to check whether a given number is prime or composite.

Methodology followed:

```
#include <iostream>
using namespace std;

int main()
{
    int num;
    cin >> num;
    bool flag = 0;

    // Check for divisors from 2 to num/2
    for (int i = 2; i < (int)num / 2; i++)
    {
        if (num % i == 0)
        {
            flag = 1;
        }
    }
```

```
    if (flag == 1)
    {
        cout << "The number is composite" << endl;
    }
    else
    {
        cout << "The number is prime" << endl;
    }
    return 0;
}
```

Input and Output:

```
PS C:\Work\comp\OOPpractical\Experiment_2> cd "c:\Work\comp\OOPpractical\Experiment_2\" ; if ($?) { g++ d2.cpp -o d2 } ; if ($?) { .\d2 }
13
The number is prime
PS C:\Work\comp\OOPpractical\Experiment_2> cd "c:\Work\comp\OOPpractical\Experiment_2\" ; if ($?) { g++ d2.cpp -o d2 } ; if ($?) { .\d2 }
17
The number is prime
PS C:\Work\comp\OOPpractical\Experiment_2> cd "c:\Work\comp\OOPpractical\Experiment_2\" ; if ($?) { g++ d2.cpp -o d2 } ; if ($?) { .\d2 }
32
The number is composite
PS C:\Work\comp\OOPpractical\Experiment_2> cd "c:\Work\comp\OOPpractical\Experiment_2\" ; if ($?) { g++ d2.cpp -o d2 } ; if ($?) { .\d2 }
88
The number is composite
PS C:\Work\comp\OOPpractical\Experiment_2>
```

Practical 2(e)

Aim: Write a program to print all Armstrong numbers between 1 and 1000.

Methodology followed:

```
// Program to find Armstrong numbers from 1 to 1000
// Armstrong: sum of digits^(digit count) equals the number
#include <iostream>
#include <string>
#include <math.h>
using namespace std;

int main()
{
    string num;
    for (int j = 1; j <= 1000; j++)
    {
        num = to_string(j);
        int sum = 0;
        int n;

        // Calculate sum of each digit raised to power of digit count
```

```
        for (int i = 0; i < num.length(); i++)
        {
            n = num[i] - '0';
            sum += pow(n, num.length());
        }

        if (sum == j)
        {
            cout << j << " is an Armstrong number" <<
endl;
        }
    }
    return 0;
}
```

Input and Output:

```
PS C:\Work\comp\OOPpractical\Experiment_2> cd "c:\Work\comp\OOPpractical\Experiment_2\" ; if ($?) { g++ e2.cpp -o e2 } ; if ($?) { .\e2 }
1 is an Armstrong number
2 is an Armstrong number
3 is an Armstrong number
4 is an Armstrong number
5 is an Armstrong number
6 is an Armstrong number
7 is an Armstrong number
8 is an Armstrong number
9 is an Armstrong number
370 is an Armstrong number
371 is an Armstrong number
407 is an Armstrong number
PS C:\Work\comp\OOPpractical\Experiment_2>
```

Practical 2(f)

Aim: Write a program that prints numbers from 1 to 100, skipping multiples of 5using a while loop and continue statement.

Methodology followed:

```
// Program to print numbers 1-100 not divisible by 5
#include <iostream>
using namespace std;

int main()
{
    int i = 1;
    while (i <= 100)
    {
        if (i % 5 == 0)
        {
            i++;
            continue;
        }
        else
        {
            cout << i << endl;
```

```
            i++;
        }
    }
    return 0;
}
```

Input and Output:

```
PS C:\Work\comp\OOPpractical\Experiment_2> cd "c:\Work\comp\OOPpractical\Experiment_2\" ; if ($?) { g++ f2.cpp -o f2 } ; if ($?) { .\f2 }
1
2
3
4
6
7
8
9
11
12
13
14
16
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18
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96
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98
99
PS C:\Work\comp\OOPpractical\Experiment_2>
```